Empirical Software Engineering manuscript No.
(will be inserted by the editor)

Posterior Guarantees Under Empirical Acquisition Cost: A Replication and Extension of FSE 2026

Authors omitted for draft

Abstract Sampling is central to performance analysis of configurable software systems: the full configuration space is often too large to measure exhaustively, so researchers and practitioners select a finite sample and reason from the observed behavior. A recent FSE 2026 paper proposed a posterior-guarantee framework for this setting. Instead of asking only whether a predictive model is accurate, it asks whether the worst observed performance bound in a sample is trustworthy, using posterior risk, posterior confidence, integrated confidence discrepancy, and Statistical Recursive Search (SRS) [WMS26]. In parallel, recent work on sampling cost argues that the usual proxy for acquisition effort, sample size, is often not the same as the empirical cost of measuring the sampled configurations [CostSamplingICSE].

This paper connects these two lines of work. We first replicate the FSE posterior-guarantee framework on the restored FSE artifact systems. We then extend the analysis to x264, energy/performance measurements, workload-sensitive configurable systems, Linux/TuxKConfig measured pools, and LCBench measured hyperparameter-configuration pools. Across 936 primary context-budget scopes, posterior risk distinguishes sampling strategies in 902 scopes, showing that the FSE metric transfers well. However, replacing nominal sample size by measured acquisition cost changes the confidence/cost Pareto frontier in 910 scopes. SRS remains strong in the direct FSE revisit and especially in x264, but it is not a universal empirical-cost-normalized recommendation: it is best in 54/102 x264 scopes, but 10/224 ENERGY scopes, 30/372 WORKLOAD scopes, 0/21 Linux scopes, and 0/210 LCBench scopes.

The contribution is therefore a refinement rather than a contradiction of FSE 2026. Posterior risk and confidence are useful, portable guarantee metrics, but the recommended sampling strategy depends on the measured acquisition cost and the dataset context. Sample size remains a useful nominal budget variable; it is not, by itself, a calibrated cost model for posterior performance guarantees.

Keywords: configurable software systems, performance sampling, posterior risk, posterior confidence, sampling cost, empirical cost, replication, SRS.

Authors omitted for review

Keywords configurable software systems · performance sampling · posterior risk · posterior confidence · sampling cost · empirical cost · replication · SRS

1 Introduction

Configurable software systems expose large configuration spaces. A compiler, a database, a video encoder, or an operating-system kernel can have tens, hundreds, or thousands of valid options, and the performance impact of those options can vary by workload, input, and measured property. Exhaustively measuring every configuration is often impractical. The usual response is to sample: select a subset of configurations, measure them, and use the sample to guide modeling, optimization, or risk assessment.

Sampling creates a second problem. Once we have measured only a subset, what can we say about the configurations that remain unmeasured? The FSE 2026 paper answers this question with a posterior-guarantee framework [WMS26]. Given a sampled performance bound, such as the worst performance value observed in the sample, the framework estimates the posterior risk that unmeasured configurations violate that bound. It also derives posterior confidence curves and an integrated confidence discrepancy that compare the observed posterior guarantee with a prior confidence baseline. The same paper proposes Statistical Recursive Search (SRS), a sampling strategy designed to discover influential decisions and improve the observed bound.

The FSE contribution is important because it changes the question from "did our model predict well?" to "how much risk remains behind the bound we observed?". That question is directly relevant when a practitioner uses sampling to make a safety, performance, or resource-planning decision. If the sample never observes a slow configuration, a large energy consumption, or a high-cost build, the practitioner needs to know whether this absence is informative or merely lucky.

At the same time, the sampling literature commonly treats sample size as the cost of a study. If two strategies both measure n configurations, they are often considered to have the same acquisition cost. This assumption is convenient and sometimes defensible for abstract analysis, but it is not generally true for performance measurement. Individual configurations can differ substantially in runtime, energy use, compilation time, storage footprint, or benchmarking duration. Two samples with the same number of configurations can require very different empirical effort. Recent cost-validity work makes this point for accuracy-based sampling studies: sample size is a nominal budget, not always a calibrated empirical cost [CostSamplingICSE].

This paper combines these two ideas as a two-paper replication. From FSE 2026, we replicate posterior risk, posterior confidence, integrated confidence discrepancy, and SRS. From the ICSE cost-validity study, we replicate the construct-validity question: whether sample size is a safe proxy for empirical sampling cost. We ask whether the FSE posterior-guarantee results replicate and transfer, and whether the FSE metrics lead to the same recommendations when empirical acquisition cost replaces sample size. The goal is not to claim that the FSE framework is wrong. On the contrary, our results show that posterior risk and posterior confidence are useful across a wider set of systems and datasets than the original paper. The goal is to test the cost model behind the recommendation. If posterior guarantees are plotted, ranked, or selected

against sample size, do the same conclusions hold when the x-axis is the measured cost actually paid to obtain the sample?

We study six primary corpora. The first is the restored FSE artifact, which provides the closest direct replication of the original setting. The second is x264 from the ICPE configurable-systems corpus, used as a positive configurable-system replication with multiple inputs and target properties [PereiraICPE20]. The third is an energy/performance corpus, which changes the measured property used as acquisition cost [TwinsEnergy]. The fourth is a workload corpus, which tests input and workload sensitivity [WorkloadICSE23]. The fifth is a Linux/TuxKConfig measured-pool stress test [TuxKConfig]. The sixth is LCBench, a hyperparameter-configuration benchmark used as a cross-domain measured-pool stress test [LCBench].

Across these corpora, the main pattern is stable. Posterior risk distinguishes strategies in almost all scopes. This supports the FSE framework. However, empirical cost changes the confidence/cost Pareto frontier in almost all scopes. This supports the cost-validity concern. SRS remains a strong and often defensible strategy in the direct FSE artifact and x264, but its advantage does not generalize uniformly to ENERGY, WORKLOAD, Linux, or LCBench once empirical cost is used.

This paper makes four contributions:

1. It provides a direct, artifact-based replication of the FSE posterior-risk framework on the restored FSE sampling data, including a documented comparison between the paper-text prior exponent and the artifact-compatible `n + 1` exponent.
2. It extends FSE-style posterior risk and confidence to five additional empirical settings: x264, ENERGY, WORKLOAD, Linux measured pools, and LCBench measured pools.
3. It evaluates the FSE metrics under measured empirical acquisition cost, showing that the cost-aware confidence Pareto frontier changes in 910/936 primary scopes.
4. It audits the local SRS implementation against the original FSE notebook cells on controlled fixtures and documents the one deliberate difference: the local implementation fills from unselected configurations so cost accounting does not double-count duplicate rows.

The rest of the paper is organized as follows. Section 2 gives background on posterior guarantees and empirical acquisition cost. Section 3 states the research questions. Section 4 describes the protocol, datasets, metrics, implementation, and statistical analysis. Sections 5 and 6 present the results for replication/transfer and empirical cost. Section 7 reports sensitivities and audits. Section 8 discusses implications. Section 9 covers threats to validity. Section 10 summarizes related work and Section 11 concludes.

2 Background and Motivation

2.1 Sampling for Configurable Systems

Configurable systems can expose a large number of valid configurations. Each configuration is a vector of decisions, and each decision can affect performance, energy, output size, compilation time, memory footprint, or other non-functional properties. The

full measurement matrix is often unavailable: measuring every configuration under every workload or input can be too slow, too expensive, or physically impractical.

Sampling strategies choose which configurations to measure. Common strategies include random sampling, pairwise or t-wise combinatorial sampling, distance-based sampling, solver-based sampling, diversity-based sampling, and adaptive strategies that use information from previous measurements. The standard evaluation question is often accuracy-based: for a fixed number of measurements, which strategy supports the most accurate model or decision?

The posterior-guarantee perspective asks a different question. Suppose the sample observes a worst-case bound, such as the maximum runtime in the sampled configurations. A model might predict many values, but a practitioner may simply need to know how credible the observed bound is. If 100 configurations were measured and none exceeded a given bound, how much risk remains that an unmeasured configuration does exceed it?

2.2 FSE Posterior Risk and Confidence

Let `C` be a finite reference set of configurations and let `f(c)` be the measured property of configuration `c`. For a max-bound analysis, a sample `S` observes the bound:

```
b(S) = max { f(c) | c in S }.
```

The posterior risk of the sample is the fraction of configurations in the reference set whose measured value exceeds that sampled bound:

```
risk(S) = | { c in C | f(c) > b(S) } | / |C|.
```

For a min-bound analysis, the inequality is reversed. The present primary protocol follows the FSE max-bound framing, with a documented lower-bound sensitivity for throughput-like WORKLOAD cases.

Posterior confidence is the empirical distribution function of posterior risks over repeated samples. For an accepted risk threshold `epsilon`, posterior confidence is the fraction of repeated samples whose posterior risk is at most `epsilon`:

```
confidence_post(epsilon) = Pr[ risk(S) <= epsilon ].
```

The FSE framework compares this posterior confidence curve with a prior confidence curve from scenario-theory reasoning. In the paper-first protocol we use the paper-text exponent:

```
confidence_prior(epsilon, n) = 1 - (1 - epsilon)^n,
```

where `n` is the nominal sample size. The released artifact also contains an `n + 1` convention; we rerun that convention as an appendix sensitivity.

The integrated posterior confidence is the area under the posterior confidence curve. For the empirical distribution used here, this is exactly:

```
area_post = 1 - mean(risk).
```

The integrated confidence discrepancy is the posterior area minus the prior area. In the rerun tables this appears as `omega_size_baseline` when the prior baseline uses

nominal sample size, and as `omega_cost_baseline` when the prior baseline uses an effective sample size induced by measured acquisition cost.

2.3 SRS

Statistical Recursive Search (SRS) is the adaptive strategy proposed in the FSE artifact. It repeatedly draws a small random batch, identifies decisions shared by high-performing configurations, tests whether those decisions are statistically associated with performance, and recursively restricts the search space. The artifact uses a default batch step size of 25 and Welch's t-test at `p < 0.05`. If recursion stops before the requested sample size, the sample is supplemented with random configurations.

Our implementation follows these defaults in `cost_sampling/srs.py`. The direct FSE artifact rerun uses the artifact-provided `SRSSampling_t2` traces rather than regenerating those samples. For generated or pool-based corpora, the local SRS implementation generates samples with the same defaults and 100 seeds where the wrapper supports repeated stochastic runs.

2.4 Sample Size as a Cost Model

A sampling study needs a cost model. The simplest model is the count model: one measured configuration costs one unit, so a sample of size `n` costs `n`. This model is attractive because it is easy to compare across systems, independent of hardware, and aligned with many theoretical treatments.

The count model is not the same as empirical acquisition cost. If one configuration takes 0.1 seconds to measure and another takes 100 seconds, a sample of 20 configurations can be cheap or expensive depending on which configurations are selected. The same issue appears when cost is energy, encoded output size, cumulative training time, compilation time, or another measured property.

This distinction matters for posterior guarantees. A guarantee obtained from 64 cheap configurations and a guarantee obtained from 64 expensive configurations may have the same nominal sample size, but they did not require the same measurement effort. A cost-aware decision should therefore consider the measured acquisition cost behind the guarantee.

2.5 Connecting the Two Lines of Work

The FSE framework supplies a useful outcome: posterior risk and posterior confidence. The cost-validity argument supplies a different x-axis: measured acquisition cost rather than nominal sample size. Combining them asks:

`Which strategy gives the best posterior guarantee for the empirical cost paid?`

This question is not answered by either line of work alone. A posterior-risk study can rank strategies by fixed sample size without checking whether fixed sample size is fixed cost. A cost-validity study can show that sample size is miscalibrated without

asking whether the resulting guarantee is trustworthy. The combined question is the focus of this paper.

3 Research Questions

We organize the journal version around two main research questions.

3.1 RQ1: Replication and Transfer of FSE Posterior Claims

RQ1. To what extent do the FSE posterior-risk results replicate on the original artifact systems and transfer to other datasets?

This RQ evaluates the posterior-guarantee framework itself. It asks whether posterior risk distinguishes strategies, whether guarantees improve with nominal budget, whether SRS remains strong, and whether these conclusions are context-sensitive.

We decompose RQ1 into four sub-questions:

- **RQ1.1 Metric transfer:** Do posterior-risk distributions distinguish sampling strategies at the same nominal budget?
- **RQ1.2 Budget response:** Do posterior guarantees improve as the nominal sample budget grows?
- **RQ1.3 SRS recommendation:** Does SRS remain a strong or best strategy under posterior-risk and posterior-confidence outcomes?
- **RQ1.4 Context sensitivity:** Where do the answers change across systems, inputs, workloads, properties, and measured pools?

3.2 RQ2: Replication of FSE Metrics Under Empirical Cost

RQ2. Do FSE posterior metrics and recommendations replicate when measured acquisition cost replaces sample size?

This RQ evaluates the cost model behind the recommendation. It adapts the existing cost-validity RQs to the posterior-guarantee setting. Prediction error is replaced by posterior risk and integrated confidence; sample-size cost is compared with measured empirical acquisition cost.

We decompose RQ2 into five sub-questions:

- **RQ2.1 Cost heterogeneity:** At fixed sample size, do strategies induce different measured acquisition costs?
- **RQ2.2 Sample-size proxy validity:** Is sample size a calibrated proxy for posterior-guarantee acquisition cost?
- **RQ2.3 Recommendation stability:** Does the top guarantee/cost strategy change when empirical cost replaces sample size?
- **RQ2.4 Pareto-front stability:** Does the confidence/cost Pareto frontier change when empirical cost replaces sample size?
- **RQ2.5 Context sensitivity:** Are cost-aware conclusions stable across datasets, properties, inputs, workloads, and measured pools?

Table 1 makes the two replication targets explicit. The first target provides the guarantee outcome and SRS treatment. The second target provides the cost-model threat and the sample-size-vs-measured-cost analysis pattern.

Table 1 Replication targets and how they are used in this study.

Target	Role in this paper
FSE 2026	Replicate posterior risk, posterior confidence, integrated discrepancy, SRS, and FSE-style claims; then test transfer beyond the released artifact.
ICSE cost study	Replicate the sample-size versus measured-cost threat model; replace prediction error with posterior guarantees.
Joint extension	Analyze sampling decisions in the measured guarantee/cost plane using aggregate counts, block tests, and curated frontier examples.

4 Experimental Protocol

4.1 Units of Analysis

The main descriptive unit is a scope: one corpus-specific context at one budget level. A context can be a system, an input/target pair, a system/workload/property tuple, a Linux release, or an LCBench dataset, depending on the corpus. A budget level is a nominal sample size such as `t2`, `n32`, or a fraction of a measured Linux pool.

For each scope, we compute posterior risks over repeated runs of each strategy. We then summarize:

- posterior-risk distributions by strategy;
- median acquisition cost and percent of full measured cost;
- effective sample size induced by measured cost;
- integrated posterior confidence;
- integrated confidence discrepancy against the size baseline;
- integrated confidence discrepancy against the measured-cost baseline;
- count-cost and measured-cost Pareto frontiers.

4.2 Corpora

Table 2 summarizes the primary corpora and their intended roles.

Table 2 Primary corpora, budgets, measured cost axis, and replication role.

Corpus	Ctx.	Budgets	Cost	Role
Original FSE artifact	7	t2	performance	Direct FSE artifact revisit.
x264	34	t1–t3	time	Configurable-system replication.
ENERGY	56	n8–n64	energy/perf.	Property/cost extension.
WORKLOAD	93	n8–n64	time	Workload/input sensitivity.
Linux full	7	1–5%	vmlinux	Measured-pool stress test.
LCBench	35	n8–n64	time_total	HPO measured-pool stress test.

The direct FSE artifact is the closest replication of the original study. The official Zenodo `data.zip` was restored and checksum-verified with SHA-256:

`12ee2295d3d9d596eb57a92008b27562a0bf1882b21fcad9c8cd82082bf654bd`

The Linux and LCBench studies are intentionally phrased as measured-pool analyses. Their posterior risk is reference-relative to the measured pool; we do not claim exhaustive risk over all possible Linux or hyperparameter configurations.

4.3 Strategies

The direct FSE artifact run uses the five paper-aligned strategies available in the restored sampling traces:

- `random`
- `solverBased`
- `divDistBased`
- `twise`
- `SRSSampling`

The extended corpora use the strategy sets supported by their runners. These include random, balanced, diversity, solver/distance or generated baselines, pilot-cost strategies where available, and SRS. Diagnostic cheap/expensive oracles are excluded from the primary generated-corpus reruns because they are local diagnostic baselines rather than FSE paper strategies.

4.4 Posterior Metrics

For each repeated sample, we compute the sampled bound and posterior risk over the reference universe. For each strategy and level, we aggregate posterior risks over runs and compute integrated posterior confidence:

`area_post = 1 - mean(risk).`

We compare this area with two prior baselines. The size baseline uses the nominal sample size `n`. The cost baseline converts the median percent of full measured acquisition cost into an effective sample size:

`effective_n_by_cost = (sample_cost / full_cost) * |C|.`

The cost-aware confidence discrepancy is then:

`omega_cost = area_post - area_prior(effective_n_by_cost).`

This makes the cost-aware comparison structurally parallel to the original size-aware comparison. It does not simply divide confidence by time; it asks whether the posterior confidence exceeds the prior confidence expected at the empirical cost actually paid.

4.5 Decision Metrics

For each scope, we compute six operational hypotheses:

Hyp.	Short question	Type
FSE1	Risk differs by strategy.	Inferential
FSE2	Empirical cost changes the top choice.	Decision
FSE3	SRS remains best under empirical cost.	Decision
FSE4	Empirical cost changes the frontier.	Decision
FSE5	SRS risk is lower than random.	Inferential
FSE6	SRS and random costs differ.	Inferential

FSE1, FSE5, and FSE6 are inferential scope-level tests. FSE2, FSE3, and FSE4 are deterministic decision indicators computed from the resulting summaries and frontiers.

4.6 Statistical Policy

The statistical layer post-processes the clean rerun outputs without modifying the raw posterior records. It applies Holm-Bonferroni correction to the scope-level inferential hypotheses FSE1, FSE5, and FSE6 within each corpus and hypothesis. Decision hypotheses FSE2, FSE3, and FSE4 are reported as indicators, not p-value tests.

For paper-level claims, repeated budget levels and stochastic runs should not be treated as independent systems. We therefore collapse scope rows into corpus-specific blocks:

- original FSE artifact: system;
- x264: input and target property;
- ENERGY: system, target property, and cost property;
- WORKLOAD: system, workload, target property, and cost property;
- Linux: release;
- LCBench: OpenML dataset.

For each corpus and hypothesis, the block majority supports or refutes the hypothesis. Mixed blocks are reported separately and omitted from the exact sign test denominator. Block-level p-values are Holm-corrected within corpus across FSE1-FSE6.

4.7 Protocol Constants and Adaptations

The paper-first reruns use these constants:

- prior confidence exponent: paper-text `n`, implemented as `--prior-offset 0`;
- SRS step size: 25;
- SRS test: Welch t-test with `p < 0.05`;
- SRS completion: random fill to requested sample size;
- repetitions: 100 stochastic/SRS seeds where the wrapper supports them;
- primary diagnostic oracles: excluded;
- primary bound orientation: max;
- primary clean reruns: table-first, with curated plots generated post hoc.

The artifact-compatible `n + 1` prior exponent is retained as an appendix sensitivity. Linux diversity sampling is the most expensive part of the full Linux run; the completed full run uses two 256-candidate diversity runs per eligible release/level while keeping random/balanced and SRS at 100 seeds. The earlier one-run, 128-candidate Linux diversity pass is retained as a sampler budget sensitivity.

5 RQ1 Results: Replication and Transfer

5.1 Primary Descriptive Results

Tables 4 and 5 summarize the primary clean reruns. A scope is one context at one budget level. The descriptive totals are useful as compact magnitude summaries, but they should not be interpreted as pooled inferential tests because corpora differ in size and structure.

Table 4 RQ1-oriented descriptive outcomes under the clean rerun protocol.

Corpus	Scopes	Risk diff.	SRS best	SRS Pareto	SRS lower risk
FSE artifact	7	7/7	3/7	6/7	6/7
x264	102	102/102	54/102	71/102	68/102
ENERGY	224	224/224	10/224	81/224	54/224
WORKLOAD	372	353/372	30/372	135/372	94/372
Linux full	21	21/21	0/21	15/21	5/21
LCBench	210	195/210	0/210	26/210	1/210
Total	936	902/936	97/936	334/936	228/936

Table 5 RQ2-oriented descriptive outcomes when measured cost replaces sample count.

Corpus	Scopes	Top changes	Frontier changes	SRS cost diff.
FSE artifact	7	2/7	6/7	7/7
x264	102	7/102	91/102	66/102
ENERGY	224	51/224	224/224	106/224
WORKLOAD	372	72/372	359/372	156/372
Linux full	21	10/21	21/21	20/21
LCBench	210	133/210	209/210	26/210
Total	936	275/936	910/936	381/936

5.2 RQ1.1: Metric Transfer

Posterior risk transfers well. Strategy posterior-risk distributions differ in 902/936 primary scopes. The direct FSE artifact, x264, ENERGY, and Linux show differences in every scope. WORKLOAD and LCBench are slightly less uniform but still high at 353/372 and 195/210 scopes.

The block-level statistical layer supports the same conclusion for the larger corpora. x264, ENERGY, WORKLOAD, and LCBench all support FSE1 at the block level after correction. The direct FSE artifact and Linux each have only seven blocks, so even

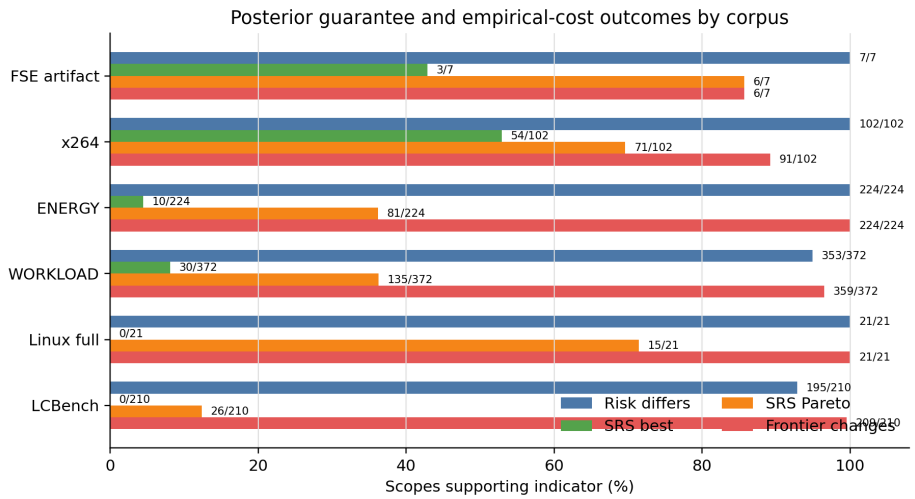


Fig. 1 Aggregate posterior-guarantee and empirical-cost outcomes by corpus. The figure mirrors the table but emphasizes the contrast between metric transfer, SRS dominance, SRS Pareto membership, and frontier change.

all-directional scope results often remain block-sign inconclusive after Holm correction. This should be read as a power and blocking limitation, not as contradictory evidence.

Answer to RQ1.1. Yes. FSE-style posterior risk is a portable outcome for comparing sampling strategies. It distinguishes strategies across the direct artifact, configurable-system replications, generated-sampling extensions, and measured-pool stress tests.

5.3 RQ1.2: Budget Response

Budget response is clearest in the corpora with comparable repeated budget levels. x264 is the cleanest example: for SRS, posterior risk is non-increasing in 34/34 input/target contexts and integrated posterior confidence is non-decreasing in 34/34 contexts. Linux also shows monotonic SRS posterior-risk improvement in the completed full pass.

This result should be interpreted carefully. A larger nominal sample usually improves the posterior guarantee, but it does not imply that sample size is a calibrated empirical cost. Two samples at the same nominal level can have different measured costs, and a strategy can improve posterior confidence while spending substantially more or less acquisition effort.

Answer to RQ1.2. Mostly yes where budget scaling is directly comparable. More samples generally improve posterior guarantees, but the improvement should not be conflated with equal empirical cost.

5.4 RQ1.3: SRS Recommendation

SRS remains important, but not universally dominant.

In the direct FSE artifact, SRS is best under the measured-cost baseline in 3/7 systems and appears on the empirical cost/confidence Pareto frontier in 6/7. This supports the original FSE intuition that SRS is a strong strategy in the artifact setting.

x264 is the strongest positive external replication. SRS is best in 54/102 scopes and appears on the empirical Pareto frontier in 71/102. It has lower posterior risk than random in 68/102 scopes. At the block level, x264 supports the SRS-risk advantage over random and is inconclusive, rather than negative, on the stricter "SRS remains best under empirical cost" question.

The other corpora weaken a universal SRS recommendation. SRS is best in only 10/224 ENERGY scopes and 30/372 WORKLOAD scopes. It is never the best cost-normalized strategy in the Linux full run or LCBench. SRS remains Pareto-efficient in many Linux scopes, but it is rarely Pareto-efficient in LCBench.

Table 6 compresses the block-level verdicts; the heatmap figure shows the full corpus-by-hypothesis grid.

Table 6 Block-level statistical verdicts, summarized by corpus.

Corpus	Block-level summary
FSE artifact	Directional but inconclusive after blocking.
x264	Supports risk transfer, SRS-risk advantage, cost differences, and frontier changes.
ENERGY	Supports risk transfer/frontier changes; refutes universal SRS best.
WORKLOAD	Supports risk transfer/frontier changes; refutes universal SRS best.
Linux full	Directional but inconclusive after seven release blocks.
LCBench	Supports risk transfer/frontier changes; refutes universal SRS best.

The "refutes" label is hypothesis-specific. For FSE3, it means the block-level evidence refutes the claim that SRS remains the best empirical-cost-normalized strategy. For FSE2, it means top-choice changes are not consistently frequent in that corpus. For FSE5, it means the evidence does not support a broad SRS posterior-risk advantage over random.

The direct FSE artifact deserves separate visual inspection because it is the closest replication target. The next figure shows how much the best measured-cost strategy improves over SRS in each original artifact system. Green bars indicate systems where SRS is still the cost-normalized winner; orange bars indicate systems where SRS remains on the measured-cost frontier but is not best. An asterisk marks systems where empirical cost changes the top recommendation.

Answer to RQ1.3. SRS remains a strong baseline and often a defensible choice, especially for the direct FSE artifact and x264. It should not be presented as a universally best cost-aware recommendation.

5.5 RQ1.4: Context Sensitivity

The posterior framework transfers more robustly than the SRS recommendation. x264 is favorable to SRS. ENERGY and WORKLOAD preserve the usefulness of

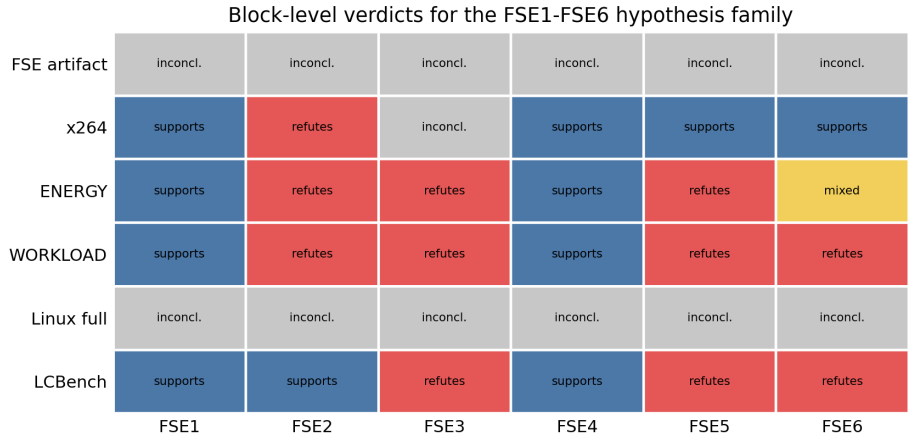


Fig. 2 Block-level verdict heatmap for FSE1-FSE6. This figure follows the FSE-style hypothesis family while using ICSE-style blocking so repeated budgets do not become pseudo-replications.

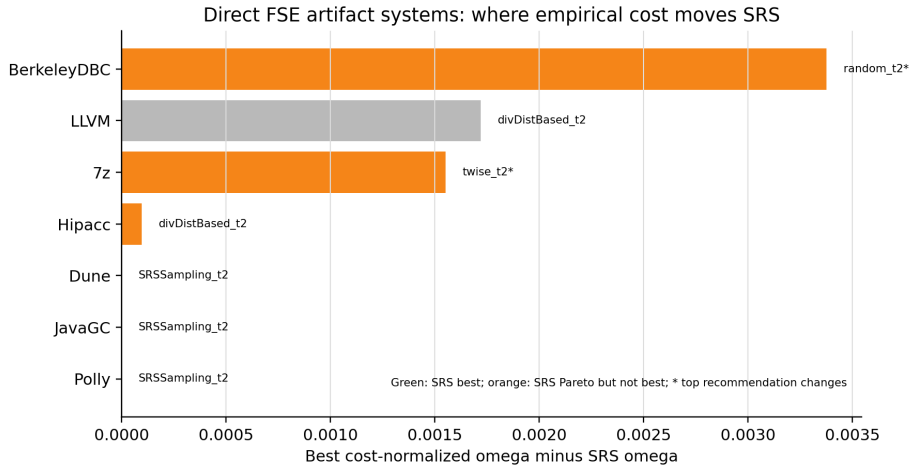


Fig. 3 Direct FSE artifact systems: measured-cost gap between the best strategy and SRS. This is the system-level companion to the aggregate SRS-best counts.

the posterior-risk metric but substantially reduce the frequency with which SRS is the best empirical-cost-normalized strategy. Linux shows SRS often remains Pareto-efficient but not top-ranked. LCBench is the strongest negative stress-test case for SRS.

This is the central context-sensitivity result. The FSE posterior-risk metric generalizes; a universal strategy recommendation does not.

The next figure gives FSE-style posterior-risk panels for three representative artifact systems. These panels preserve the original paper's visual language: they compare strategies by posterior risk at the fixed artifact budget before the paper turns to the measured-cost extension.

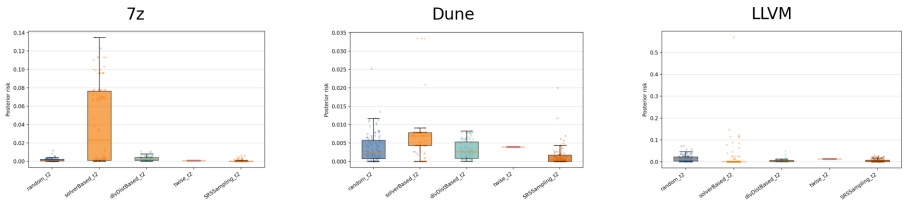


Fig. 4 FSE-style posterior-risk panels for representative original artifact systems. These panels anchor the extension in the replicated FSE outcome before empirical acquisition cost is introduced.

Answer to RQ1.4. Context matters. The guarantee framework transfers across contexts, but the strategy recommendation depends on the corpus, workload, property, and acquisition-cost model.

6 RQ2 Results: Empirical Acquisition Cost

6.1 RQ2.1: Cost Heterogeneity

At fixed nominal sample size, strategies often induce different measured acquisition costs. SRS and random costs differ in 381/936 descriptive scopes. The rate is especially high in the direct FSE artifact (7/7), x264 (66/102), and Linux (20/21).

The lower rates in ENERGY, WORKLOAD, and LCBench should not be read as evidence that sample size is generally safe. These corpora still show extensive frontier changes, meaning that even when a particular pairwise cost comparison is not consistently decisive, the overall guarantee/cost decision surface can change.

Answer to RQ2.1. Fixed sample size does not generally imply fixed measured acquisition cost. Cost heterogeneity is present and strategy-dependent.

6.2 RQ2.2: Sample-Size Proxy Validity

The posterior rerun evaluates sample-size proxy validity by asking whether the scientific decisions agree under two baselines:

- the size baseline, where prior confidence is computed using nominal `n`;
- the cost baseline, where prior confidence is computed using the effective sample size implied by measured acquisition cost.

If sample size were a safe proxy, the top recommendations and Pareto frontiers would usually agree. They do not. The top recommendation changes in 275/936 descriptive scopes, and the confidence/cost Pareto frontier changes in 910/936.

Top changes are context-dependent; Pareto changes are nearly universal. This is important. A top-1 winner is a scalar summary that can remain stable even when many defensible tradeoff options change. The Pareto frontier exposes the structure of the decision surface. Under empirical acquisition cost, that surface is usually not the same as under nominal sample size.

Answer to RQ2.2. Sample size is a useful nominal budget parameter, but it is not a calibrated proxy for empirical acquisition cost when making posterior-guarantee decisions.

6.3 RQ2.3: Recommendation Stability

The single top recommendation changes in some, but not all, corpora:

- direct FSE artifact: 2/7 scopes;
- x264: 7/102 scopes;
- ENERGY: 51/224 scopes;
- WORKLOAD: 72/372 scopes;
- Linux: 10/21 scopes;
- LCBench: 133/210 scopes.

LCBench is the strongest top-change result. At the block level, LCBench supports frequent top recommendation changes. x264, ENERGY, and WORKLOAD refute frequent top changes at the block level, even though many individual scopes change. The direct FSE artifact and Linux are small-block and inconclusive.

The conclusion is therefore not that empirical cost always changes the top choice. Instead, empirical cost sometimes changes the top choice, and the frequency of that change is itself context-sensitive.

Answer to RQ2.3. Empirical cost can change the top recommendation, but top-1 instability is not the most general result. It is strongest in LCBench and visible in Linux and the direct artifact; it is less frequent in x264, ENERGY, and WORKLOAD.

6.4 RQ2.4: Pareto-Front Stability

Pareto-front stability is the strongest empirical-cost result. The measured cost/confidence frontier changes in 910/936 descriptive scopes:

- direct FSE artifact: 6/7;
- x264: 91/102;
- ENERGY: 224/224;
- WORKLOAD: 359/372;
- Linux: 21/21;
- LCBench: 209/210.

The block-level synthesis supports frontier changes for x264, ENERGY, WORKLOAD, and LCBench. Direct FSE and Linux are directionally strong but inconclusive at the corrected block-sign level because each has only seven blocks.

This result is more important for practitioners than a top-1 flip alone. A Pareto frontier represents the choices that are not obviously wasteful: no other available strategy/level pair offers at least as much confidence for no more cost. If empirical cost changes that set, then a practitioner using sample-size cost may discard a defensible measured-cost choice or retain a choice that is dominated once real acquisition cost is considered.

Answer to RQ2.4. Yes. Replacing sample size with measured acquisition cost changes the confidence/cost Pareto frontier in almost every primary scope.

6.5 RQ2.5: Cost-Aware Context Sensitivity

The cost-aware conclusions differ by corpus.

x264 remains favorable to SRS: SRS is often best and often Pareto-efficient. Nevertheless, empirical-cost frontiers change in 91/102 x264 scopes. This means the top recommendation can remain stable while the surrounding set of defensible choices changes.

ENERGY and WORKLOAD show that the posterior metrics transfer, but SRS is not often the best measured-cost choice. These corpora are important because they remain within configurable-system style settings while changing measured properties and workload contexts.

Linux shows a different pattern. SRS is never the best cost-normalized strategy in the completed full run, but it remains on the measured-cost Pareto frontier in 15/21 scopes. That is a nuanced negative result: SRS is not dominant, but it is often not wasteful.

LCBench is the strongest external stress test. SRS is never the best cost-normalized strategy and is Pareto-efficient in only 26/210 scopes. This does not refute FSE for configurable software; it shows that the SRS recommendation should not be generalized without checking domain and cost structure.

Answer to RQ2.5. The effect of empirical cost is context-dependent. The most stable result is not a universal winner change, but a widespread change in the guarantee/cost Pareto frontier.

6.6 Curated Frontier Examples

The curated frontier example set consists of five panels generated by the curated-frontier artifact. These panels illustrate the main decision patterns without dumping every per-context plot.

Panel A: the restored FSE 7z system. SRS remains Pareto-efficient, but measured cost changes the top recommendation to `twise_t2`.

Panel B: x264 time input 1. SRS is both the measured-cost winner and Pareto-efficient.

Panel C: WORKLOAD kanzi/vmlinux. SRS is Pareto-efficient but not the best measured-cost option.

Panel D: Linux 5.0.0 measured pool. SRS is Pareto-efficient but not dominant.

Panel E: LCBench mfeat-factors. SRS is not on the measured-cost frontier.

For journal layout, these panels can be combined into one multi-panel figure with a shared legend and a caption explaining that the x-axis is measured acquisition cost and the y-axis is integrated posterior confidence.

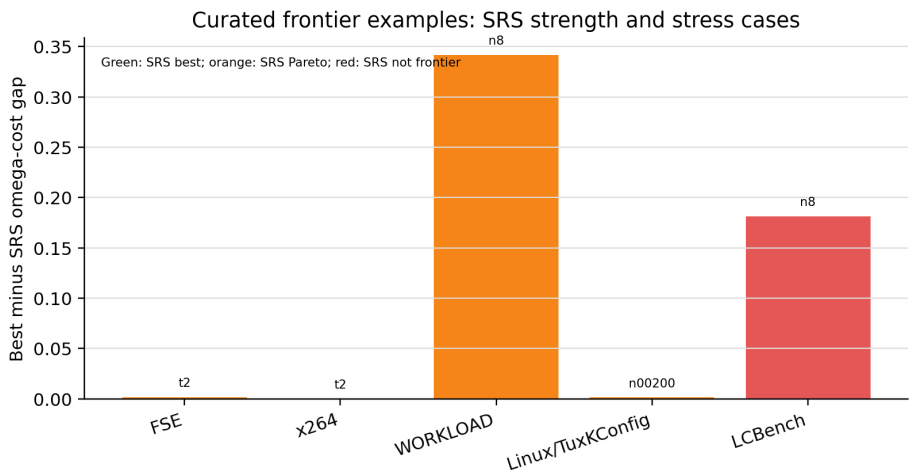


Fig. 5 Summary of the five curated frontier examples. The bars show the best-minus-SRS omega-cost gap; green means SRS is best, orange means SRS is on the frontier but not best, and red means SRS is not on the measured-cost frontier.

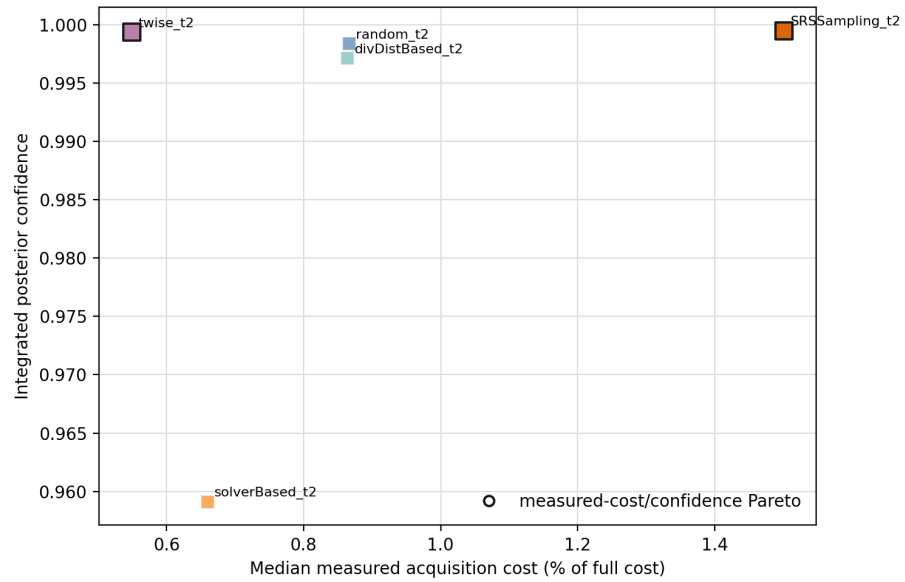


Fig. 6 Direct FSE 7z frontier

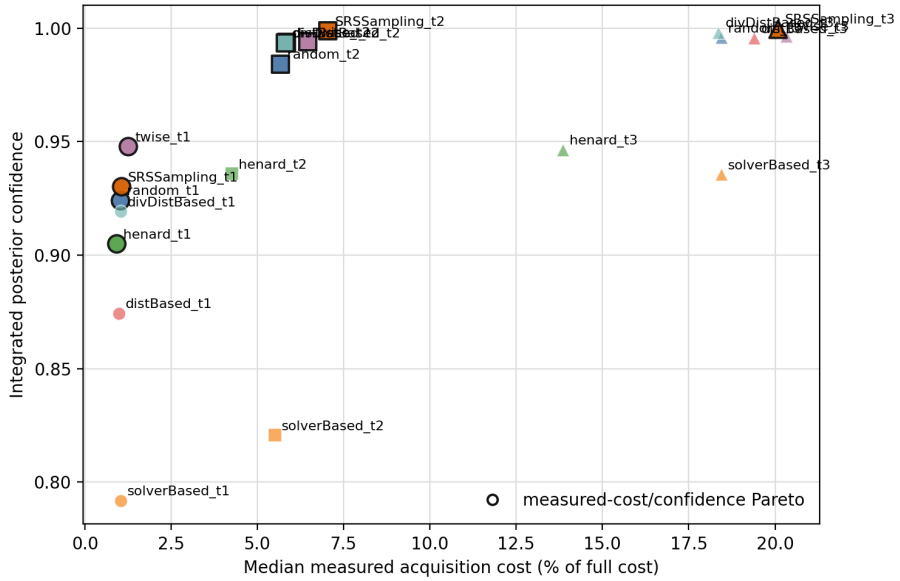


Fig. 7 x264 frontier

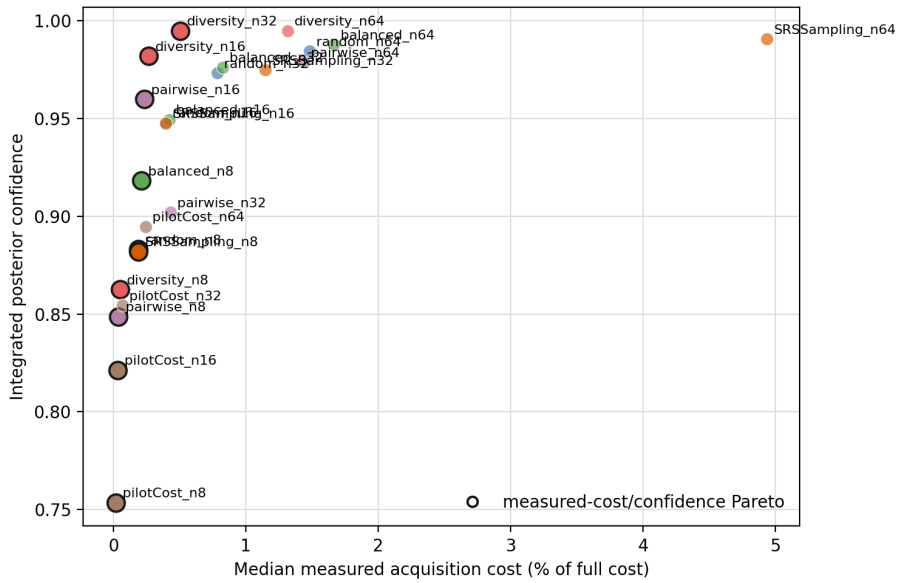


Fig. 8 WORKLOAD frontier

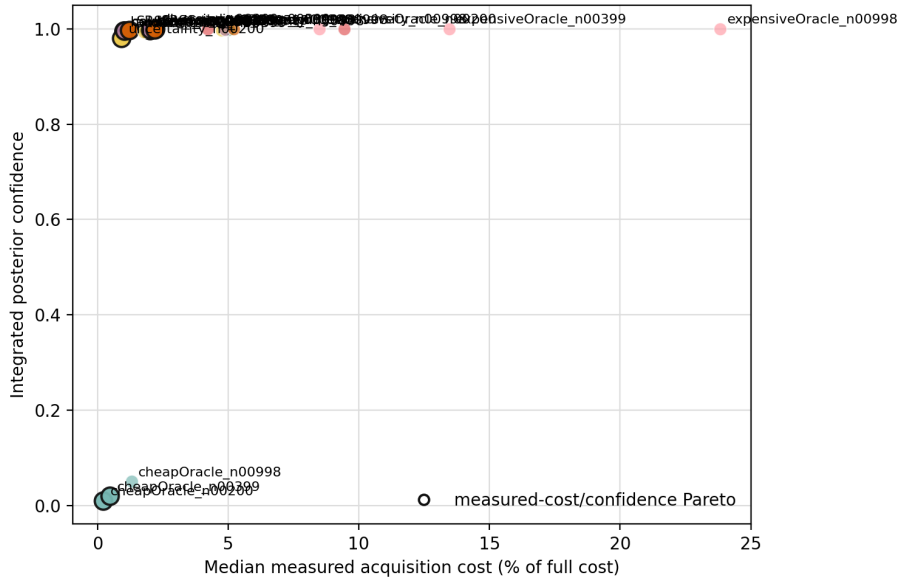


Fig. 9 Linux frontier

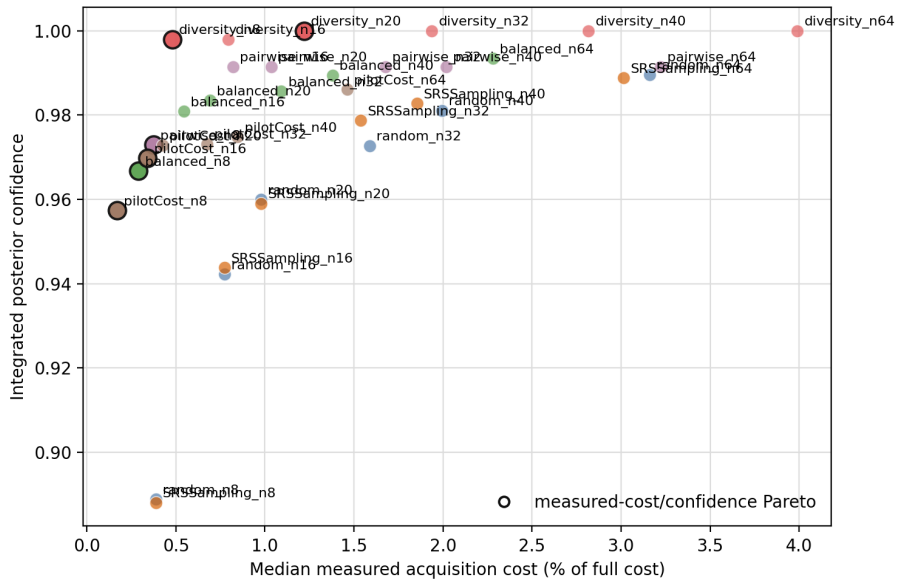


Fig. 10 LCBench frontier

7 Sensitivities and Audits

7.1 Prior Exponent Sensitivity

The paper-first protocol uses the paper-text prior exponent `n`. The released artifact convention uses `n + 1`. We reran the direct FSE artifact with the artifact-compatible exponent as an appendix sensitivity.

The headline counts are identical:

- SRS best: 3/7;
- SRS Pareto: 6/7;
- top recommendation changes: 2/7;
- frontier changes: 6/7.

This means the main conclusion is not driven by the exponent convention. The `n + 1` run should remain in the appendix, because the paper-text definition is the clearer main-text choice and the sensitivity validates robustness.

7.2 x264 Size as Cost

The x264 primary run uses encoding time as acquisition cost while evaluating time and size targets. We also reran x264 with encoded size as acquisition cost. SRS remains favorable:

- SRS best: 57/102;
- SRS Pareto: 76/102;
- frontier changes: 96/102.

This supports the claim that x264 is a positive external replication for SRS under both primary cost choices, while still preserving the cost-aware frontier result.

7.3 WORKLOAD Throughput Framing

The primary WORKLOAD run treats `h2` throughput as an inverse acquisition-cost quantity. We also ran an `h2` lower-bound framing sensitivity. In that sensitivity:

- SRS best: 1/32;
- SRS Pareto: 5/32;
- frontier changes: 31/32.

This sensitivity does not make SRS dominant. It supports moving detailed throughput-orientation discussion to the appendix.

7.4 LCBench Cumulative Runtime

The primary LCBench run uses the last observed runtime mode. A cumulative runtime sensitivity gives the same headline counts:

- SRS best: 0/210;

- SRS Pareto: 26/210;
- frontier changes: 209/210.

The LCBench stress-test conclusion is robust to this runtime-mode choice.

7.5 Linux Diversity Budget

The completed Linux full run uses two 256-candidate diversity runs per eligible release/level. An earlier one-run, 128-candidate diversity pass is retained as a sampler-budget sensitivity. It preserves the key conclusions:

- SRS best: 0/21;
- SRS Pareto: 15/21;
- frontier changes: 21/21.

This suggests that increasing the diversity candidate budget strengthens some inferential details but does not change the SRS-best or Pareto conclusions.

7.6 SRS Notebook Replay Audit

The local SRS implementation is a code-native implementation aligned with the FSE artifact notebook. The preserved notebook snapshot is:

`artifacts/srs_original_fse_implementation/SRS-sampling.ipynb`
`SHA-256: 24769e71e5827a98278cbb148ba1268d4d446d11fb5816fe62955d3994e250f0`

The notebook replay audit executes the preserved notebook import/helper cells on controlled fixtures and compares row outputs with `cost_sampling.srs`. Three cases are recorded:

- a single-batch fixture with exact row identity match;
- a recursive strong-signal fixture with exact row identity match;
- a duplicate-fill fixture with the expected documented divergence.

The divergence is intentional. The notebook may fill from the original measurement pool and repeat rows already selected. The local implementation fills from remaining rows so acquisition-cost accounting does not double-count a configuration. For strict bit-for-bit historical replay, the audit harness should be run in the original artifact's pandas/SciPy environment. For the present paper, the correct wording is "faithful code-native implementation with controlled output-level replay," not "bit-for-bit artifact reuse under all library versions."

8 Discussion

8.1 A Replication, Not a Rejection

The results should be read as a replication-plus-extension of FSE 2026. The posterior-risk metric works. It distinguishes strategies across many corpora, and it gives a meaningful guarantee-oriented view that is different from prediction error. The direct artifact revisit also shows that SRS remains strong in the original setting.

The extension is about cost. The FSE recommendation is naturally stated at a fixed nominal sample size. We ask whether the same recommendation holds when the acquisition cost is the measured effort required to obtain the sample. The answer is nuanced: often yes for x264, sometimes for the direct artifact, but not generally across all external settings.

8.2 Why Pareto Frontiers Matter

Top-1 recommendations are easy to report but brittle. A single scalar winner depends on the objective, normalization, and ties. Pareto frontiers are more decision-oriented. They show which choices are not dominated in the confidence/cost plane.

The strongest empirical-cost result is that Pareto frontiers change in 910/936 primary scopes. This means empirical cost changes the decision set even when the single top choice remains stable. For practitioners, this is more useful than simply asking whether SRS wins. It identifies which strategy/budget pairs remain defensible once the actual measurement cost is considered.

8.3 SRS as a Strong Baseline, Not a Universal Winner

The paper should avoid overstating the negative result. SRS is not "bad." It is often Pareto-efficient and sometimes best. In x264, it is the strongest positive external result. In Linux, it is never the top cost-normalized choice but remains Pareto-efficient in most scopes. In the direct FSE artifact, it is Pareto-efficient in 6/7 systems.

The correct conclusion is that SRS should be treated as a strong baseline whose cost-aware value depends on the dataset and cost axis. A practitioner should not assume that SRS is the best empirical-cost strategy without measuring the cost profile of the candidate samples.

8.4 Effective Cost Baselines Are More Informative Than Raw Ratios Alone

The cost-aware posterior metric used here is not merely "confidence divided by cost." It converts measured acquisition cost into an effective sample size and then asks whether the observed posterior confidence exceeds the prior confidence expected at that effective budget. This keeps the comparison close to the FSE framework while replacing the cost abstraction.

Raw calibration ratios remain useful for diagnosing cost heterogeneity, but the posterior question is ultimately decision-oriented: which choices remain defensible in the confidence/cost plane?

8.5 Practical Implications

For researchers, the implication is methodological. Reporting posterior guarantees only at fixed sample sizes can hide cost-model assumptions. Future sampling studies

should report both nominal size and measured acquisition cost when the measured property is heterogeneous.

For practitioners, the implication is operational. A sampling strategy should be selected not only by how many configurations it measures, but by the guarantee obtained for the measured cost spent. A strategy that samples fewer configurations is not necessarily cheaper; a strategy that samples the same number of configurations is not necessarily equal cost.

For artifact evaluators, the implication is reproducibility-oriented. The direct FSE rerun required restoring the official artifact data and documenting the prior-exponent convention. The SRS audit required preserving the original notebook and comparing outputs on controlled fixtures. These details matter when a replication shifts from metric reproduction to cost-aware extension.

9 Threats to Validity

9.1 Construct Validity

The main construct threat is the definition of acquisition cost. Different corpora expose different measured properties. In x264, primary acquisition cost is encoding time and appendix cost is encoded size. In ENERGY, cost can be energy or performance. In WORKLOAD, cost is time except for throughput-like cases. In Linux, the measured pool uses `vmlinux` as the cost-relevant quantity. In LCBench, cost is `time_total`.

This heterogeneity is intentional because the research question is about measured acquisition cost, not one universal cost unit. Still, the final paper must be explicit that cost is corpus-specific and that cross-corpus totals are descriptive, not a single pooled cost scale.

A second construct threat is the use of measured-pool posterior risk for Linux and LCBench. These are not exhaustive configuration spaces. We address this by using "pool-relative" or "reference-relative" posterior risk for those corpora.

9.2 Internal Validity

The posterior implementation was written locally and aligned with the FSE artifact formulas. The prior exponent differs between the paper text and the artifact notebook; we make the paper-text `n` convention primary and rerun the artifact-compatible `n + 1` convention as a sensitivity. The sensitivity does not change the direct artifact headline counts.

The SRS implementation is code-native. Controlled notebook replay shows exact matches on two fixtures and a documented duplicate-fill divergence on one fixture. The direct FSE artifact run uses the artifact's shipped SRS traces, reducing risk for the direct replication. Generated and measured-pool corpora use the local implementation.

Linux diversity sampling required a runtime adaptation. The completed full run uses two 256-candidate diversity runs per eligible level; the earlier one-run 128-candidate pass is retained as a sensitivity and preserves the key SRS-best and Pareto conclusions.

9.3 Statistical Conclusion Validity

Scope-level rows are not all independent. A corpus can contribute many budget levels, inputs, or repeated stochastic runs. We therefore use block-level sign tests for paper-level claims, with blocks defined by systems, input/target pairs, workload contexts, releases, or datasets as appropriate.

The block-level tests are conservative for small-block corpora. The direct FSE artifact and Linux each have only seven blocks, so strong descriptive counts can remain statistically inconclusive after Holm correction. We report both scope-level magnitudes and block-level verdicts to avoid overclaiming.

9.4 External Validity

The six corpora are broad but not exhaustive. The direct FSE artifact covers the original setting. x264, ENERGY, and WORKLOAD broaden configurable-system evidence. Linux and LCBench provide measured-pool stress tests. Other systems, hardware platforms, workloads, and measurement protocols may yield different cost structures.

The result should therefore be generalized as a methodological caution: posterior guarantees should be evaluated against measured cost when acquisition cost is heterogeneous. It should not be generalized as a claim that one particular alternative strategy always dominates SRS.

9.5 Reproducibility

All primary and appendix reruns are recorded as artifact directories in this repository. The clean protocol, compact synthesis, statistical tests, curated frontiers, SRS notebook audit, and implementation modules are part of the artifact package.

The restored FSE data archive is checksum-verified, but some raw external inputs, such as LCBench JSON and workload archives, remain local external data dependencies. Final artifact packaging should either provide those inputs, document download instructions, or provide checksums and expected paths.

10 Related Work

10.1 Posterior Guarantees for Performance Sampling

The closest related work is the FSE 2026 posterior-risk paper that introduced the guarantee framework and SRS [WMS26]. Our work directly replicates that framework on the restored artifact and then extends it to measured empirical acquisition cost. The present study should therefore be positioned as a replication and cost-aware extension, not as an unrelated competing method.

10.2 Sampling Strategies for Configurable Systems

Configurable-system sampling has studied random, combinatorial, distance-based, solver-based, diversity-based, and adaptive strategies [KalteneckerICSE19, PereiraICPE20]. Much of this work uses sample size as the primary cost axis. The x264 and broader configurable-system corpora used here inherit that tradition, but we reinterpret their samples through measured cost and posterior guarantees.

10.3 Empirical Cost of Sampling

The cost-validity line of work argues that sample size is not necessarily sampling cost [CostSamplingICSE]. It studies whether conclusions based on count-cost differ from conclusions based on measured acquisition time, energy, or related properties. Earlier cost-aware configurable-systems work also considered measurement effort, but often through count-based abstractions such as fixed train/test acquisition counts [SarkarASE15]. This paper reuses the empirical-cost framing but changes the outcome from prediction error to posterior risk and confidence.

10.4 Workload, Input, and Property Sensitivity

Prior configurable-system studies show that performance behavior can depend on inputs, workloads, and non-functional properties [PereiraICPE20, WorkloadICSE23, TwinsEnergy]. Our WORKLOAD and ENERGY extensions are aligned with that literature. The novelty here is the cost-aware posterior-guarantee question: not merely whether behavior changes across contexts, but whether the posterior guarantee/cost decision surface changes.

10.5 Hyperparameter and Measured-Pool Benchmarks

LCBench and Linux/TuxKConfig measured pools are not direct replicas of the FSE configuration-space setting [LCBench, TuxKConfig]. They are stress tests for the methodology. They ask whether the posterior-guarantee and empirical-cost reasoning behaves similarly in large measured pools with heterogeneous acquisition costs. This helps delimit the scope of the SRS recommendation.

11 Artifact and Data Availability

The current draft is backed by the following committed artifact groups:

- protocol and synthesis reports;
- statistical-testing outputs;
- curated frontier figures and manifest;
- SRS notebook audit and notebook replay notes;
- SRS and posterior implementation modules.

Primary rerun directories:

- direct FSE artifact clean paper protocol;
- x264 ICPE paper protocol;
- ENERGY paper protocol;
- WORKLOAD paper protocol;
- Linux full diversity protocol;
- LCBench 100-seed paper protocol.

Appendix rerun directories:

- direct FSE artifact `n + 1` sensitivity;
- x264 size-cost sensitivity;
- WORKLOAD throughput sensitivity;
- LCBench cumulative-runtime sensitivity;
- Linux full default-diversity sensitivity.

Before submission, this section should be converted into the journal's artifact availability format, with persistent archive identifiers and checksums for all external raw inputs.

12 Conclusion

This paper revisits FSE 2026 posterior performance guarantees under a measured acquisition-cost model. The result is a refinement of the FSE story. Posterior risk and posterior confidence replicate and transfer well: they distinguish sampling strategies in 902/936 primary scopes. SRS remains a strong strategy in the direct FSE artifact and x264. However, sample size is not a reliable proxy for empirical acquisition cost when making guarantee/cost decisions. Replacing nominal sample size with measured cost changes the confidence/cost Pareto frontier in 910/936 primary scopes, and SRS is not the universal cost-normalized winner across ENERGY, WORKLOAD, Linux, and LCBench.

The practical message is straightforward. Posterior guarantees answer an important question: how much risk remains behind a sampled bound? To make that answer actionable, the guarantee should be interpreted against the empirical cost paid to obtain it. Sample size remains useful for experiment design, but measured acquisition cost is the relevant budget for cost-aware decisions.

13 Draft Reference Keys

The current repository does not contain a BibTeX file. The following draft keys identify the references that need bibliographic metadata before LaTeX conversion or journal submission.

- `[WMS26]` FSE 2026 posterior-risk and SRS paper under replication.
- `[CostSamplingICSE]` "Sample Size Is Not Sampling Cost" ICSE submission and public reproducibility repository.
- `[KalteneckerICSE19]` Configurable-system sampling corpus used in the REPLICATION family.
- `[PereiraICPE20]` x264 ICPE configurable-system artifact and dataset.
- `[TwinsEnergy]` Twins/Twins-or-False-Friends energy/performance corpus.

745 – [WorkloadICSE23] Workload-impact configurable-system corpus.
746 – [TuxKConfig] Linux/TuxKConfig measured-pool data.
747 – [LCBench] LCBench lightweight benchmark data.
748 – [SarkarASE15] Prior cost-aware configurable-systems sampling work using a
749 count-based cost abstraction.